Digital Access: AltNarrative, a multilingual digital repository, and a Comics Studies Lab for born-digital comics

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Dennis Kogel contends that non-text contents are made accessible to the visually impaired by use of online tools such as "alternative text" or "alt text" (23-24). [1] He further elaborates that these 'alt text' not only "display an alternative description to the image" but also "give additional information" (24). [1] He also saw web cartoonists "consciously subverting the intended use of alt text" by adding "jokes" and "additional information for the strip" and playing "with the expectations of what to usually find in alt text" (25). [1] Building on this subversive function of alt text, Frank Bramlett argues and demonstrates how "alt-text" creates "an authorial presence" that results "in a 'metacomic' effect" (72). [2] According to Makayla Lewis, Miriam Sturdee, John Miers, Josh Urban Davis, and Thuong Hoang these "typical alt text" which "simply describe what was seen in the image" do not "work on sketches". [3] To overcome this limitation, they introduce a new tool called "AltNarrative" that "describes the concept portrayed by the sketches". [3]

StoryWeaver is "a digital repository of multilingual stories for children from *Pratham Books*" in India. [4] *StoryWeaver* takes "books to some of the most disadvantaged children globally, particularly where chronic under-investment in the availability of mother tongue reading resources poses a significant barrier to literacy". [4] Its story page also states:

Every book is freely available in multiple formats can be read online, and keeping in mind the digital divide - can also be **read offline, downloaded, printed** and even repurposed, so that no child is left behind. **Translation and versioning tools help customize the books for localized requirements** and these resources become available to other users as well, creating a multiplier effect and amplifying impact. [4]

This platform also adopts an 'open' philosophy in its objective of "creating book security for every child". [4]

At the Indian Institute of Technology Jodhpur, I plan to set up a Comics Studies Lab with the aim to make comics accessible to all. The first output of this lab is to create an inclusive digital comic book which is in an unconventional format being simultaneously immersive, interactive, multidimensional, multilingual, multimedial, multimodal, and multisensory. Toward this end, the technical plan is to create a comic book that can be read in a virtual reality environment using an ocular device enhanced with assistive technologies such as haptic devices to experience the sense of touch and for mobile accessibility to physically impaired; artificial touch for immersive virtual reality; multilingual devices, natural language processing devices, and translation devices for linguistic accessibility; corrective lenses and eEye for visual accessibility to visually impaired; hearing/auditory devices for auditory impairment; eNose and artificial smell to experience the sense of smell; and electric taste simulation technology and eTongue to experience the sense of taste.

The Web Content Accessibility Guidelines 2.2 provides usability guidelines which are organized under the four principles of perceivability, operability, understandability, and robustness. These guidelines, also self-announces that, they "address problem particular to people with disabilities". [5]

The first narrative brings together scholars from UK, US, and Australian universities working at the intersections of HCI and comics. The second narrative is told from the perspectives of " a not-for-profit children's book publisher" in India. The final narrative is a scholarly project from an academic institution in India. These three narratives are invoked to demonstrate disruptive digital projects that attempt to make comics accessible to all be it through AltNarrative, a multilingual digital repository, or the inception of a lab that will create a born-digital and an inclusive comic of the future. This paper thus critically examines the three narratives interrogating the issue of web or digital accessibility in them while also evaluating these web contents for compliance with WCAG2.2.

1. Kogel, Dennis. 2013. "Rethinking Webcomics: Webcomics as a Screen Based Medium." Unpublished MA thesis. University of Jyväskylä, Finland.
2. Bramlett, Frank. 2018. Linguistic discourse in webcomics: Extending conversation and nar-

rative into alt-Text and hidden comics. In *The Language of Pop Culture* . Routledge, 72–91.
3. Lewis, M., Sturdee, M., Miers, J., Davis, J. U., & Hoang, T. (2022, April). Exploring AltNarrative in HCI imagery and comics. In *CHI conference on human factors in computing systems extended abstracts* (pp. 1–13). https://dl.acm.org/doi/10.1145/3491101.3516394
4. https://storyweaver.org.in/en/about/our-story
5. https://www.w3.org/WAI/WCAG22/Understanding/intro#understanding-the-four-principles-of-accessibility